20/01/2026. Tutorial-3.

a A mobile service provider wants to classify SMS mes as Spam or Not Spam using a single-layer neural network in the layer.

X_1 = Specific words and phrases.

X_2 = Links and Attachments

X_3 = Bias.

X_1	X_2	X_3	Target (Y)
4	2	1	1
16	0	1	0
6	3	1	1
2	1	1	0

Hidden layer

Inputs	H.L	Output
X_1	h_1	$v_1 = 0.1$
X_2	h_2	$v_2 = 0.2$
X_3	h_3	$v_3 = 0.3$
	h_4	$v_4 = 0.4$
	h_5	$v_5 = 0.3$

	w_1	w_2	b
h_1	0.1	0.2	1
h_2	-0.2	0.3	1
h_3	0.4	-0.1	1
h_4	0.3	0.2	1
h_5	0.2	-0.4	1

h1

$z_1 = ?$

$X_1 = 4$ $X_2 = 2$ $b = 1$ $w_1 = 0.1$ $w_2 = 0.2$

$z_1 = 0.4 + 0.4 + 1$

$\boxed{z_1 = 1.8}$ $\boxed{y = 1}$

h2

$z_2 = ?$ $X_1 = 4$ $X_2 = 2$ $b = 1$

$w = -0.2$ $w_2 = 0.3$

$\boxed{z = 0.8}$ $\boxed{y = 1}$

Quick Work

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M T W T F S S

h3

$z_3 = ?$ $x_1 = 4$ $x_2 = 2$ $b = 1$ $w_1 = 0.4$ $w_2 = -0.1$

$\boxed{z_3 = 2.4}$ $\boxed{\hat{y} = 1}$

h4

$z_4 = ?$

$x_1 = 4$ $x_2 = 2$ $b = 1$ $w_1 = 0.3$ $w_2 = 0.2$

$\boxed{z_4 = 2.6}$ $\boxed{\hat{y} = 1}$

h5

$z_5 = ?$

$x_1 = 0,\ x_2 = 0,\ b = 1$

$\boxed{z_5 = 1}$

$$\underbrace{h_1 = \sigma(z_1)}_{h_1} = \frac{1}{(1+e^{-z})} = \frac{1}{1+e^{-1.8}} = 0.86$$

$$\underbrace{\sigma(z_2)}_{h_2} = \frac{1}{1+e^{-0.8}} = 0.69$$

$$\underbrace{\sigma(z_3)}_{h_3} = \frac{1}{1+e^{-2.4}} = 0.92$$

$$\underbrace{\sigma(z_4)}_{h_4} = \frac{1}{1+e^{-2.6}} = 0.93$$

$$\underbrace{\sigma(z_5)}_{h_5} = \frac{1}{1+e^{-1}} = 0.73.$$

final: $z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5 = ?$

$= (0.86)(0.1) + (0.69)(0.2) + (0.92)(0.3) + (0.93)(0.4) + (0.73)(-0.3)$

$= 0.377 \approx 0.38.$

a $\sigma(z_{out}) = \frac{1}{(1+e^{-0.38})} = 0.59 \approx 0.60$

$$\hat{y} = \begin{cases} 0 & \sigma(z) < 0.5 \\ 1 & \sigma(z) \geq 0.5 \end{cases}$$

$\hat{y}$ (or) y predicted

$$w_i^{New} = w_i^{old} + \eta(y - \hat{y}) h_i$$

η → Learning rate

y → True

$\hat{y}$ → predict

$$w_i^{New} = w_i^{old} + \eta(y - \hat{y}) x_i$$